Analysis of Weather Simulation Algorithms Using Asymptotic Notation

SOURCE CODE:

```
#include <stdio.h>
#include <math.h>
int main()
{
    int t, i, n;
    printf("Enter the number of input sizes: ");
    scanf("%d", &t);
    printf("\n----------------------------------------------------------------------\n");
    printf("Input\t\tO(n(log n)^2)\t\tO(n^2 log n)\n");
    printf("----------------------------------------------------------------------\n");
    for(i = 1; i <= t; i++)
    {
        printf("\nEnter input size %d: ", i);
        scanf("%d", &n);
        double logn = log(n) / log(2);
        double complexity1 = n * logn * logn;      // O(n(log n)^2)
        double complexity2 = n * n * logn;         // O(n^2 log n)
        printf("%d\t\t%.2f\t\t\t%.2f\n", n, complexity1, complexity2);
    }
    printf("\n----------------------------------------------------------------------\n");
    printf("\nConclusion:\n");
    printf("O(n(log n)^2) grows much slower than O(n^2 log n).\n");
    printf("Therefore, O(n(log n)^2) is more scalable and efficient for\n");
    printf("large-scale weather simulations.\n");
```

```
    return 0;
}
```